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Investigation on the key factors associated with flavor quality in northern strong aroma type of Baijiu by flavor matrix

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ABSTRACT

The strong aroma type of Baijiu is popular with its balanced flavor. However, the flavor characteristics of strong aroma type of Baijiu in different regions have significant diversities. Among them, the key aroma compounds in northern strong aroma type of Baijiu and the factors affecting the aroma profile and flavor quality are still unclear. In this study, a total of 78 aroma compounds were identified by gas chromatography–olfactometry-mass spectrometry (GC–O–MS). The quantitative analysis was completed and ethyl hexanoate (1003.24–4506.04 mg/L) was the highest. Aroma profile was successfully simulated by reconstitution, and omission test was applied to investigate the effects of the important aroma compounds on the aroma profile. Additionally, the relationship between expression of aroma compounds and sensory characteristics of samples was illustrated by flavor matrix. The overall aroma profile of northern strong aroma type of Baijiu was formed by the key aroma compounds with different aroma expression.

1. Introduction

Baijiu distilled from the fermented grains is widely recognized as the national liquor of China because of its complex production process and popular flavor characteristics. Based on the differences in the aroma profiles, Baijiu can be roughly classified into 12 types, in which strong aroma, sauce aroma, and light aroma are deemed as 3 basic aroma types because of their popular flavor characteristics. In particular, strong aroma type of Baijiu is popular with customers for its much more balanced flavor, which occupies a momentous position in the Baijiu market. In the past five years, the market share of strong aroma type of Baijiu always had been around 50%, far ahead of other aroma types of Baijiu.

For the production process of strong aroma type of Baijiu, sorghum, wheat, barley, corn and other grains are crushed and mixed with Jiuqu as starter and saccharifying agent, and then put into the pit for fermentation. Afterwards, Jiupei (fermented grains) are distilled in the Zeng (a unique container) to obtain the fresh Baijiu. Due to the complex raw materials, special technology and co-fermentation of multiple strains (Hu, Du, & Xu, 2015; Liu et al., 2017), the trace components in strong aroma type of Baijiu are also abundant. As previously reported (Sun, 2019), 861 trace components were found in strong aroma type of Baijiu, including 248 esters, 13 lactones, 122 alcohols, 65 acids, 53 aldehydes, 83 ketones, 35 carbonyls, 63 hydrocarbons, 24 phenols, 16 ethers, 29 sulfur compounds, 71 nitrogen compounds, 35 furans, and 4 others.

The flavor characteristics of alcoholic beverages are deemed as important factors affecting their quality and consumer acceptance. (Yu, Nie, Chen, & Xu, 2023) Certainly, the flavor quality of strong aroma type of Baijiu mainly depends on the rich trace components, and the brewing environment and related factors have significant impacts on the formation of these trace components. (Hong, Zhao, & Sun, 2021; Wu et al. 2023; Zhu et al., 2020; Fan & Qian, 2005; Xu et al., 2023; Mao et al., 2022) Thus, even for the same aroma type, the flavor characteristics of strong aroma type of Baijiu distributed in different regions also have significant diversities. (He, Liu, Qian, Yu, Xu & Chen, 2020; Hong et al., 2021; Song et al., 2020) According to flavor characteristics and regional distribution, strong aroma type of Baijiu can also be divided into

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Sichuan category, Jianghuai category and North category. Most of the famous strong aroma type of Baijiu are produced in Sichuan province and the Huaihe River basin because of their sufficient water resources for the convenience of Baijiu making. Moreover, due to the influence of brewing environment and related factors, the pit aroma of Sichuan category is more intense, and the aroma profile of Jianghuai category is mellow and moderate. However, with the diversified demands of consumers and the improvement of production technology, strong aroma type of Baijiu in northern China has also received more and more attention in recent years. Compared with strong aroma type of Baijiu of the other two regions, there are few studies on the aroma compounds responsible for aroma profiles of strong aroma type of Baijiu in northern China. Additionally, in the actual production process, unstable flavor quality still exists in northern strong aroma type of Baijiu, and the influencing factors are unclear. There are also few studies on the correlation between key aroma compounds and flavor quality for northern strong aroma type of Baijiu.

In this study, the objectives were (1) to identify and analyze the potential aroma active compounds in northern strong aroma type of Baijiu by modern flavoromics; (2) to quantitatively analyze key aroma compounds and perform recombination omission test; (3) to estimate the relationship between aroma compounds and aroma attributes by a multivariate statistical analysis and understand aroma profile of northern strong aroma type of Baijiu, and further verify the chemical origin of their sensory attributes.

2. Material and methods

2.1. Baijiu samples

A total of three grades of northern strong aroma type of Baijiu samples were used in this study, which were directly stored in pottery after fermentation and distillation, without blending. These samples were made from the same raw materials and traditional processes and produced in Chengde Qianlongzui Distillery Co., LTD, Hebei (40.59 °N 117.57 °E, China). The grades of samples were determined based on the evaluation results of professional evaluation experts. All the samples were stored at 4 °C prior to analysis.

2.2. Analytical reagents

Ethanol (99.9%, HPLC grade) was purchased from Aladdin Bio-Chem Technology Co. (Shanghai, China). Water was purified using a Milli-Q purification system (Millipore, Bedford, MA). Dichloromethane (99.9%, HPLC grade) was purchased from Aladdin Bio-Chem Technology Company (Shanghai, China). A C3–C30 *n*-alkane mixture (Sigma–Aldrich, Shanghai, China) was used for determination of linear retention indices (RIs). The analytical standards employed for the identification and quantitative analysis of the aroma compounds were obtained from J&K Scientific Company (Beijing, China) with at least 97% purity. Pentyl acetate (internal standard (IS), IS1), 4-octanol (IS2) and 2-ethylbutyric acid (IS3) were also purchased from J&K Scientific Company (Beijing, China).

2.3. Isolation of the volatile compounds by liquid–liquid extraction (LLE)

The 30 mL sample was diluted to 15% ethanol (v/v) with saturated salt water, and extracted three times with dichloromethane (50 mL each time). The isolated organic phase was then separated into acidic fractions (AF) and neutral/basic fractions (NBF) following the procedure described by the following methods. The combined extract (about 150 mL) was extracted three times by the Na_2CO_3 solution (50 mL each time; 0.50 mol/L; pH 10.0) and then washed by 50 mL of the saturated NaCl solution. The organic phase, containing the neutral/basic aroma compounds, was named as the NBF. The combined aqueous phase (about 200 mL) was acidified to pH 2.0 with HCl (4.0 mol/L), and extracted three times by the freshly distilled dichloromethane (70 mL each time). The combined extract (about 210 mL), containing the remaining acidic aroma compounds, was named as the AF. Next, being dried over anhydrous Na_2SO_4 for 12 h at −20°C, both fractions were separated and concentrated to 1 mL using a rotary evaporator. The concentrated extracts were stored at −20°C for further analysis. (Zhao et al., 2018).

2.4. Identification and analysis of aroma compounds

Gas chromatography–olfactometry-mass spectrometry (GC–O–MS) was used to determine potential aroma active compounds. Each sample was analyzed in three replicates. A polar chromatographic column DB-WAX (60 m × 0.25 mm × 0.25 μm; Agilent Technologies, Santa Clara, CA) was used as the column. Every sample (1 μL) was injected in a spitless mode and analyzed. The temperature of the olfactory port was kept at 250 °C. Three trained panelists (one male and two females) were selected for the GC–O analysis. During a gas chromatography (GC) run, each panelist placed his or her nose close to and above the top of the sniffing port, recorded the odor of the chromatographic effluent as well as retention time. All analyses were repeated in triplicate by each panelist. (Chen et al., 2022b).

Oven temperature was held at 40 °C at first, then raised to 50 °C at a rate of 10 °C/min and held for 20 min, then raised at 1 °C/min up to 70 °C and held for 10 min, finally raised at 3 °C/min up to 250 °C and held for 15 min. The mass spectrometry (MS) was operated in an electron ionization (EI) mode at 70 eV. The temperature of the interface and the ion source were set at 250 and 230 °C, respectively. The identification of aroma compounds was conducted in a full scan mode. The mass range was set from 45 to 350 amu.

In addition to applying standard comparison and comparison retention index (RI), a compound was considered to be tentatively identified if the similarity between mass spectrometric information of each chromatographic peak and the National Institute of Standards and Technology (NIST) and LIQUOR (team self-built mass spectra library) mass spectra library was at least 80%.

2.5. Odor specific magnitude estimation (Osme)

During a GC run described above, a panelist placed his/her nose close to the sniffing port, responded to the aroma intensity of the stimulus, and recorded the aroma descriptor and intensity values as well as retention time. The aroma descriptors were determined by an evaluation of the flavor quality of reference compounds previously (Zhao et al., 2018; Chen et al., 2022b; Gao, Fan & Xu, 2014; Chen et al., 2022b). Three panelists were familiar with the aroma descriptors according to the aroma standard provided. A six-point scale ranging from 0 to 5 was used for intensity judgment: 0 = none, 1 = very weak, 2 = weak, 3 = moderate, 4 = strong, and 5 = very strong. The Osme value for aroma intensity was an average result of panelists. (Fan & Xu, 2014).

2.6. Aroma extract dilution analysis (AEDA)

The concentrated extracts (AF or NBF fraction) were diluted 3-fold in a stepwise manner by dichloromethane according to the reference (Sun et al., 2017). Each compound was assigned a flavor dilution (FD) factor representing the highest diluted sample in which the panelists could detect the odorant by using GC–O–MS. The analysis of each extract was an average result of panelists. (Fan & Xu, 2014).

2.7. Quantification and odor active values (OAVs) of aroma compounds

Major aroma compounds were directly quantified using an internal standard method by GC − MS/SIM equipped with a DB-FFAP column (60 m × 0.25 mm, 0.25 μm film thickness, J&W Scientific, USA). Mixed standard solutions were prepared as follows. The stock solution was prepared in 70% ethanol/water (v/v) solution and was then diluted to a

Table 1
Aroma intensity and FD of potential aroma active compounds.

NO.	Compounds	CAS	Aroma characteristics	RI	Aroma intensity			FD		
				DB-WAX	A	B	C	A	B	C
1	2-Butanol	78-92-2	fruity	1012	2	——	——	243	——	——
2	2-Methyl-1-propanol	78-83-1	Ethereal, grass	1073	2	2	2	81	81	27
3	2-Pentanol	6032-29-7	apple	——	1	——	——	27	——	——
4	1-Butanol	71-36-3	sweet	——	2	1	2	729	243	243
5	1-Propanol	71-23-8	fruity, banana	1031	2	——	——	81	1	1
6	3-Methyl-1-butanol	123-51-3	fruity, banana	1206	3	3	2	243	243	243
7	1-Pentanol	71-41-0	soap, sweet	——	——	——	1	——	——	9
8	1-Hexanol	111-27-3	grass, grain	1358	2	2	1	81	81	81
9	1-Heptanol	111-70-6	grass	1456	1	2	——	9	9	——
10	1-Octanol	111-87-5	orange	1550	——	1	——	——	3	——
11	1,2-Propanediol	57-55-6	alcoholic	——	——	——	1	——	——	1
12	2,3-Butanediol	513-85-9	fruity	——	——	——	1	——	——	1
13	2-Methyl-1-butanol	137-32-6	fruity	1203	2	2	2	27	27	27
14	Furfural	98-01-1	sweet	——	2	3	2	243	243	9
15	2-Furanmethanol	98-00-0	rose	1635	3	2	2	81	81	243
16	(2,2-diethoxyethyl)-Benzene	6314-97-2	cream cake	1701	2	2	——	9	9	——
17	Ethyl phenylacetate	101-97-3	rose perfume	1765	2	3	1	27	9	3
18	Ethyl 3-phenylpropionate	2021-28-5	flower	1868	3	3	2	27	9	9
19	Phenethyl alcohol	60-12-8	sweet	1889	2	2	2	243	81	243
20	Phenylacetaldehyde	122-78-1	rose	1621	1	1	2	9	9	27
21	Phenol	108-95-2	plastic, rubbery	1975	1	——	——	1	——	——
22	4-Ethylguaiacol	2785-89-9	bacon, grass	——	1	——	——	1	——	——
23	4-Ethylphenol	123-07-9	smoky	2148	1	——	——	1	——	——
24	2-Butylfuran	4466-24-4	fruity	1267	——	——	1	——	——	1
25	Benzaldehyde	100-52-7	cherry	1491	——	2	2	——	1	1
26	4-Methylphenol	106-44-5	cheese	2069	3	3	1	81	27	243
27	Phenethyl acetate	103-45-7	smoky	——	——	2	3	——	9	27
28	3-Furaldehyde	498-60-2	flower	1439	1	1	——	1	1	——
29	Furfuryl acetate	623-17-6	fruity, banana	1517	1	——	——	1	——	——
30	2,4-Decadienal	2363-88-4	grain	——	——	——	2	——	——	9
31	Acetaldehyde	75-07-0	fruity	——	3	3	3	9	9	9
32	Acetic acid	64-19-7	vinegar	1428	1	1	2	27	27	81
33	Propanoic acid	79-09-4	cheesy	1514	1	1	——	3	3	——
34	2-Methylpropionic acid	79-31-2	acidic, cheesy	1547	——	1	——	——	1	——
35	Butanoic acid	107-92-6	cheesy	1600	4	4	2	243	243	81
36	3-Methylbutanoic acid	503-74-2	cheesy	1647	3	4	1	243	81	27
37	Pentanoic acid	109-52-4	cheesy	1712	4	4	2	243	729	27
38	4-Methylpentanoic acid	646-07-1	cheesy	——	1	1	——	1	1	——
39	Hexanoic acid	142-62-1	fatty, cheesy	1827	4	4	3	243	729	81
40	Heptanoic acid	111-14-8	cheesy, sweaty	1929	2	1	1	3	1	1
41	Octanoic acid	124-07-2	cheesy	2037	2	2	1	243	27	9
42	1,1,3-Triethoxy-propane	7789-92-6	vegetables	——	1	1	1	1	1	1
43	Acetal	105-57-7	vegetables	——	1	1	——	1	1	——
44	1,1-Diethoxy-3-methyl-butane	3842-03-3	fruity	——	——	1	1	——	1	1
45	1,1-Diethoxy-hexane	3658-93-3	pear	——	1	——	——	1	——	——
46	1,1-Diethoxy-nonane	54815-13-3	rose	1527	1	——	——	1	——	——
47	Ethyl acetate	141-78-6	fruity, sweet	——	2	2	2	243	729	2187
48	Ethyl propionate	105-37-3	fruity, grape	——	——	2	——	——	9	——
49	Ethyl butanoate	105-54-4	fruity	1031	3	3	2	729	729	81
50	Isobutyl acetate	110-19-0	banana	——	——	——	2	——	——	9
51	Ethyl nonanoate	123-29-5	fruity	——	1	1	——	1	1	——
52	Isoamyl acetate	123-92-2	banana	1115	——	——	2	——	——	1
53	Ethyl pentanoate	539-82-2	apple, pineapple	1129	3	3	1	729	243	27
54	Ethyl hexanoate	123-66-0	pineapple	1238	5	4	2	6561	2187	729
55	Methyl hexanoate	106-70-7	strawberry	1194	1	——	——	9	——	——
56	Hexyl acetate	142-92-7	apple	1289	2	——	1	27	——	9
57	Propyl hexanoate	626-77-7	sweet, fruity	1333	2	——	——	27	——	——
58	Ethyl lactate	97-64-3	fruity, buttery	1351	1	3	2	81	81	243
59	Ethyl heptanoate	106-30-9	pineapple	1354	3	3	2	27	27	27
60	Hexyl butanoate	2639-63-6	apple	1409	1	——	——	9	——	——
61	Butyl hexanoate	626-82-4	fruity	1409	2	2	——	243	9	——
62	Ethyl octanoate	106-32-1	banana, grape	1436	2	3	1	81	81	9
63	Butyl lactate	138-22-7	creamy	1506	2	——	——	9	——	——
64	Isobutyl lactate	585-24-0	buttery	——	——	1	——	——	1	——
65	Pentanoic acid,2-hydroxy-4-methyl-, ethyl ester	10348-47-7	fruity	——	2	2	1	9	3	3
66	Isoamyl lactate	19329-89-6	nutty	1557	——	1	——	——	9	——
67	Isopentyl hexanoate	2198-61-0	pineapple	1458	2	——	——	9	——	——
68	Hexyl hexanoate	6378-65-0	fruity	1602	2	1	——	81	1	——
69	Diethyl butanedioate	123-25-1	apple	1659	1	2	1	3	9	9
70	Ethyl decanoate	110-38-3	sweet	1630	3	——	——	9	——	——
71	Ethyl tetradecanoate	124-06-1	sweet	2045	1	——	——	9	——	——
72	Ethyl dodecanoate	106-33-2	fruity	1838	1	——	——	9	——	——
73	Ethyl hexadecanoate	628-97-7	fruity	2260	2	1	1	27	9	9

(*continued on next page*)

Table 1 (*continued*)

NO.	Compounds	CAS	Aroma characteristics	RI	Aroma intensity			FD		
				DB-WAX	A	B	C	A	B	C
74	Ethyl oleate	111-62-6	fruity	2476	1	——	——	3	——	——
75	Butyl butanoate	109-21-7	fruity	——	3	——	——	9	——	——
76	Ethyl pentadecanoate	41114-00-5	sweet	——	1	——	——	1	——	——
77	Ethyl heptadecanoate	14010-23-2	apple	——	1	——	——	1	——	——
78	Ethyl linoleate	544-35-4	fatty	2549	1	——	——	1	——	——

RI: The linear retention index on the DB-WAX column as calculated by Kovats method (Van den Dool & Kratz, 1963);
FD: The flavor dilution representing the highest diluted sample in which the panelists could detect the odorant by using GC–O–MS;
A, B, C: 3 samples from northern strong aroma type of Baijiu;
——: No identified, no found, or no calculated.

series of concentrations to obtain the working standard solutions. IS 1 ~ 3 were added to the above solutions with final concentrations of 40 mg/L, 20 mg/L, and 20 mg/L, respectively. Meanwhile, add internal standards 1–3 of the same concentration to three samples. Finally, 1.0 μL of Baijiu samples and the working standard solutions were injected into GC in splitless under the same conditions as mentioned above. Calibration curves were drawn by plotting the response ratio between the target aroma compounds and corresponding IS listed in Table 2 versus the ratio between their concentrations. The concentrations of the target aroma compounds were calculated based on the calibration curves. The analytical limits of detection (LOD) of aroma compounds were obtained from the lowest concentrations of the analyte standard solutions based on a signal-to-noise ratio of 3. All analyses were repeated in triplicate. The odor thresholds of aroma compounds were obtained from the previous papers (Liu & Sun, 2018; Fan & Xu, 2011; Wang, Chen, Wu, & Zhao, 2022). Based on the results of quantification, the OAVs of aroma compounds were calculated. (Fan & Xu, 2014; Chen et al., 2022a).

2.8. Descriptive sensory analysis (DSA)

Students and employees of the School of Light Industry at Beijing Technology and Business University were recruited as panel candidates. The general training schedule consisted of 16 h (2 h/week) for introduction to sensory analysis, aroma description and identification, ranking, and triangle tests. Sensory attributes were listed based on the description of flavor and taste of northern strong aroma type of Baijiu, including flower and fruity, alcoholic, pit aroma, grain aroma, bran aroma, dregs aroma, aged aroma, off-flavor, acidity, bitterness, sweety, spicy, and astringent. (Wang, Chen, Wu, & Zhao, 2022; He, Liu, Qian, Yu, Xu, & Chen, 2020) Candidates' performances were monitored to ensure the effectiveness of the training. 14 assessors (7 females and 7 males) with good performances were selected to form the DSA panel. Sensory attributes of samples are scored with 0–5 by DSA panel, in which 0 represents the weakest intensity, and 5 represents the strongest intensity.

2.9. Aroma recombination experiments

To validate the importance and contribution of the aromas in the 3 strong aroma type of Baijiu samples, the odorants with high OAVs (≥1) were dissolved in 3 ethanol solutions according to their detected concentrations in the 3 strong aroma type of Baijiu samples. The alcohol contents (v/v) of these ethanol solutions were 76%, 73%, and 69%, which were the same as their respective samples. The simulated samples were evaluated as described in section 2.8, and compared with their corresponding strong aroma type of Baijiu samples. (Fan & Xu, 2014).

2.10. Aroma omission experiments

The omission tests were conducted to further confirm the key odorants in the strong aroma type of Baijiu samples. A comparative analysis of the ellipsis models and the respective completed recombination models was performed using the triangle tests. The significance levels of the detected difference were calculated according to the previous method (Wang, Guo, Sun, Li, Zheng, Li, & Meng, 2022; Chen et al., 2022b).

2.11. Statistical analysis and statistical methods

According to the actual production situation and sensory evaluation results, the data set of flavor characteristics of samples was comprehensively constructed. At the same time, the data set of compounds was constructed based on the results of Osme, AEDA and OAVs. The correlation analysis of the two data sets was conducted with Pearson correlation coefficient to identify the key aroma compounds which affected flavor quality in northern strong aroma type of Baijiu.

Statistical analyses were performed using the SPSS version 22.0 Statistical Package for Windows (SPSS Inc., Chicago, Illinois, USA). The data obtained from aroma profiling and quantitative analysis were assessed by analysis of variance (ANOVA). Graphics were drawn using Photoshop 2021 (Adobe Photoshop Co., San Jose, USA) and Origin 2021 software (OriginLab Co., Northampton, MA).

3. Results and discussion

3.1. Chemical analysis

3.1.1. Identification and expression evaluation of aroma compounds

For the purpose of facilitating GC–O–MS analysis and compounds identification, the samples were further separated into acidic/water and neutral/basic fractions to reduce complexity. Following this procedure, a total of 78 aroma compounds were identified by LLE combined with GC–O–MS and Osme methods, including 2 aldehyde, 5 acetals, 10 acids, 13 alcohols, 16 aromatic compounds and 32 esters. Among them, 31 aroma compounds were detected in samples of all grades, including 1 aldehyde, 2 alcohols, 3 acetals, 6 aromatic compounds, 9 esters and 10 acids. The results showed that alcohols and esters mainly exhibited fruity and sweet aroma, while acids showed cheesy and fatty aroma. Moreover, the aroma intensities of aroma compounds were recorded according to Osme. In general, aroma compounds in sample A were the richest, and the aroma intensity of ester compounds was higher among all compounds. By comparison, fewer aroma compounds were in sample B, but the aroma intensity was similar to that of A. Meanwhile, the aroma intensity of acid compounds was higher in B. Furthermore, the number of aroma compounds in sample C was less, and their aroma intensities were generally low. In order to further confirm the contribution for aroma compounds to the flavor characteristics of northern strong aroma type of Baijiu, AEDA was then carried out.

As presented in Table 1, the FD values of those aroma compounds were shown based on the AEDA/GC–O analysis. Although their aroma intensities were not high, some of these compounds had high FD values, such as 1-butanol, ethyl acetate, etc. Significantly, 3-methyl-1-butanol, butanoic acid, hexanoic acid, ethyl hexanoate and ethyl butanoate had both higher aroma intensities and FD values in three samples. These

Table 2
The concentration and OAVs of aroma compounds in northern strong aroma type of Baijiu.

No.	Compounds	Quantified ion	IS	Standard curve			LOD (μg/L)	Concentration (mg/L)						Odor threshold (mg/L)	OAV		
				Slope	Intercept	R^2		A	RSD (%)	B	RSD (%)	C	RSD (%)		A	B	C
4	1-Butanol	56	2	0.6764	−0.6002	0.9949	33.92	329.18a	1.37%	167.03c	3.03%	237.27b	1.06%	2.73	121	61	87
6	3-Methyl-1-butanol	55	2	0.4436	5.8877	0.9952	12.72	301.36a	1.84%	294.96a	1.12%	140.31b	1.42%	179	2	2	1
5	1-Propanol	31	2	0.5794	0.2113	0.9986	26.77	463.67a	2.14%	243.96c	2.26%	273.98b	1.43%	54	9	5	5
1	2-Butanol	45	2	1.0306	0.1	0.9966	17.73	173.67	2.86%	——	——	——	——	50	3	——	——
3	2-Pentanol	45	2	1.4535	−0.0152	0.9969	27.47	35.64	1.76%	——	——	——	——	——	——	——	——
9	1-Heptanol	70	2	0.8805	−0.2619	0.998	19.61	6.79a	1.10%	6.81a	1.14%	——	——	26.6	<1	<1	——
11	1,2-Propanediol	45	2	0.9501	−0.1821	0.9932	14.95	——	——	——	——	25.33	2.98%	——	——	——	——
2	2-Methyl-1-propanol	43	2	0.0264	0.0035	0.9961	27.79	270.14a	2.11%	279.26a	1.85%	168.62b	3.95%	40	7	7	4
13	2-Methyl-1-butanol	70	2	1.7603	1.4968	0.9918	18.42	52.10b	3.10%	55.73a	2.69%	26.03c	3.17%	——	——	——	——
7	1-Pentanol	55	2	0.1674	0.0331	0.9927	5.96	——	——	——	——	35.33	1.40%	4	——	——	9
8	1-Hexanol	56	2	0.2289	0.0432	0.999	6.98	208.58a	4.93%	196.64a	1.18%	159.17b	4.61%	5.37	39	37	30
12	2,3-Butanediol	57	2	0.1562	0.0008	0.9989	7.59	68.07c	2.99%	88.24b	3.31%	174.54a	1.27%	668	<1	<1	<1
24	2-Butylfuran	81	1	0.9015	0.0401	0.9941	8.44	2.08c	1.03%	2.19b	1.57%	3.15a	1.24%	——	——	——	——
14	Furfural	96	1	3.3565	−0.1423	0.9903	5.76	2.39c	1.94%	3.21a	1.61%	2.89b	1.30%	0.122	20	26	24
15	2-Furanmethanol	98	1	0.4775	−0.0002	0.9989	3.73	3.70c	1.72%	5.25b	3.54%	10.15a	2.42%	2	2	3	5
17	Ethyl phenylacetate	91	1	3.2608	−0.1371	0.9973	9.52	2.98b	1.12%	3.94a	1.15%	2.74c	3.97%	0.407	7	10	7
19	Phenylethyl Alcohol	91	1	2.4899	0.367	0.9905	6.11	1.44c	2.96%	3.38a	4.57%	2.12b	4.59%	28.9	<1	<1	<1
18	Ethyl 3-phenylpropionate	104	1	7.5229	−0.6932	0.9976	11.13	4.23b	1.11%	4.39a	1.23%	4.04c	1.39%	0.125	34	35	32
21	Phenol	94	1	2.096	0.3342	0.9932	7.80	0.52	3.01%	——	——	——	——	18.9	<1	——	——
22	4-Ethylguaiacol	137	1	0.1028	0.007	0.9900	4.73	0.68	2.70%	——	——	——	——	0.007	98	——	——
23	4-Ethylphenol	107	1	7.153	−0.2643	0.9914	10.99	1.72	1.60%	——	——	——	——	0.618	3	——	——
25	Benzaldehyde	51	1	0.8707	−0.0333	0.9961	9.37	1.94c	1.83%	2.41b	4.56%	2.74a	5.01%	4.2	<1	1	1
27	Phenethyl acetate	104	1	5.5356	−0.3329	0.9941	7.86	——	——	2.66a	2.37%	2.55a	3.06%	0.908	——	3	3
28	3-Furaldehyde	67	1	0.9408	−0.0192	0.9999	9.11	5.79b	1.81%	11.98a	1.65%	——	——	——	——	——	——
26	4-Methylphenol	107	1	6.8948	0.0113	0.9951	7.38	0.39a	3.98%	0.21b	4.47%	0.18c	3.50%	0.167	2	1	1
20	Phenylacetaldehyde	91	1	2.5715	−0.2376	0.9957	6.77	4.44c	1.79%	4.84b	1.68%	6.46a	1.30%	0.262	17	18	25
34	2-Methylpropionic acid	73	3	1.1625	0.2163	0.9981	60.34	0.49b	3.02%	2.74a	1.39%	——	——	1.58	<1	2	——
36	3-Methylbutanoic acid	60	3	2.2907	0.4957	0.9907	36.48	11.21a	2.92%	10.88a	4.44%	1.37b	1.78%	1.045	11	10	1
32	Acetic acid	60	3	0.3059	0.0997	0.9995	59.73	121.65c	3.05%	133.47b	2.04%	364.29a	1.67%	160	1	1	2
33	Propanoic acid	74	3	0.5158	−0.5705	0.9935	54.87	24.22a	3.35%	23.88a	1.07%	——	——	18.1	1	1	——
35	Butanoic acid	60	3	1.5878	−0.0512	0.9984	70.53	41.65a	1.89%	34.42b	2.69%	21.36c	3.13%	0.964	43	36	22
37	Pentanoic acid	60	3	1.7803	−0.1207	0.9993	71.06	10.67a	2.26%	8.17b	3.07%	4.84c	4.51%	0.389	27	21	12
39	Hexanoic acid	60	3	1.3571	1.254	0.9905	66.84	250.25a	1.47%	201.44b	4.02%	92.56c	1.14%	2.52	99	80	37
40	Heptanoic acid	60	3	1.7097	0.0265	0.9915	58.18	4.01a	2.03%	0.68c	1.11%	1.23b	2.28%	13.8	<1	<1	<1
41	Octanoic acid	60	3	0.9566	−0.8439	0.9921	62.37	18.90b	1.21%	20.03a	1.53%	18.68b	1.19%	2.7	7	7	7
38	4-Methylpentanoic acid	57	3	1.0597	0.1601	0.9906	63.84	0.22a	3.22%	0.07b	1.30%	——	——	0.144	2	1	——
43	Acetal	45	1	0.4256	0.0069	0.9984	3.23	661.89a	3.81%	411.11b	2.71%	——	——	——	——	——	——
45	1,1-diethoxy-Hexane	103	1	1.9436	0.0653	0.9967	4.55	0.28a	1.23%	0.07b	1.09%	——	——	——	——	——	——
47	Ethyl acetate	43	1	0.5662	2.0779	0.9997	33.40	1752.07b	1.66%	1113.45c	1.32%	2824.67a	1.87%	32.6	54	34	87
54	Ethyl hexanoate	88	1	0.7669	−0.9888	0.9972	7.50	4506.04a	2.76%	1661.12b	1.28%	1003.24c	1.25%	0.055	81,928	30,202	18,241
66	Isoamyl lactate	45	1	0.8721	−0.0095	0.9995	6.99	——	——	9.15	1.01%	——	——	100	——	<1	——
71	Ethyl tetradecanoate	88	1	2.9565	−0.404	0.991	27.23	6.95a	2.04%	——	——	0.20b	2.23%	46.607	<1	——	<1
62	Ethyl octanoate	88	1	1.4887	−1.4681	0.991	17.00	81.61a	2.20%	76.44b	1.62%	47.89c	2.06%	0.013	6278	5880	3684
53	Ethyl pentanoate	85	1	0.7217	−0.3949	0.9906	16.34	236.26a	3.61%	93.77b	1.58%	54.68c	4.81%	0.027	8750	3473	2025
49	Ethyl butyrate	71	1	0.9468	−1.0477	0.9917	23.81	702.35a	1.13%	276.53b	2.17%	194.47c	1.69%	0.0815	8618	3393	2386
61	Butyl hexanoate	99	1	0.7646	−0.0415	0.9993	9.03	20.91a	3.80%	6.41b	3.62%	——	——	5.25	4	1	——
55	Methyl hexanoate	74	1	1.6786	−0.1212	0.993	16.52	7.48	1.87%	——	——	——	——	——	——	——	——
60	Hexyl butanoate	89	1	0.6797	−0.0267	0.9978	9.86	3.76	2.67%	——	——	——	——	——	——	——	——
75	Butyl butanoate	71	1	1.0749	−0.0503	0.9971	8.86	23.95	3.86%	——	——	——	——	0.11	218	——	——

(*continued on next page*)

Table 2 (*continued*)

No.	Compounds	Quantified ion	IS	Standard curve			LOD (μg/L)	Concentration (mg/L)						Odor threshold (mg/L)	OAV		
				Slope	Intercept	R^2		A	RSD (%)	B	RSD (%)	C	RSD (%)		A	B	C
58	Ethyl lactate	45	1	1.0573	7.293	0.9901	29.91	1145.28c	1.64%	1419.35b	2.60%	1960.62a	1.73%	128	9	11	15
59	Ethyl heptanoate	88	1	5.1922	−0.1469	0.9996	7.64	25.78a	2.99%	15.61b	1.23%	9.23c	2.44%	13.2	2	1	1
67	Isopentyl hexanoate	70	1	1.4972	0.0169	0.994	9.05	9.58	1.25%	——	——	——	——	1.4	7	——	——
65	Pentanoic acid, 2-hydroxy-4-methyl-, ethyl ester	69	1	3.5206	0.0485	0.995	6.10	5.83b	1.01%	6.65a	3.82%	4.52c	3.60%	1.268	5	5	4
68	Hexyl hexanoate	117	1	1.0521	0.0527	0.995	9.08	1.26a	1.18%	0.56b	4.83%	——	——	1.89	1	<1	——
69	Diethyl butanedioate	101	1	2.0632	−0.149	0.9928	9.29	4.93b	1.55%	7.58a	1.13%	7.82a	2.17%	353.193	<1	<1	<1
70	Ethyl decanoate	88	1	2.3026	−0.0018	0.995	8.54	0.53a	1.22%	0.36b	2.32%	——	——	1.12	<1	<1	——
72	Ethyl dodecanoate	101	1	2.3084	−0.1306	0.994	35.29	6.81	1.26%	——	——	——	——	0.4	17	——	——
76	Ethyl pentadecanoate	88	1	12.686	−0.801	0.9934	8.54	2.66	1.40%	——	——	——	——	——	——	——	——
73	Ethyl hexadecanoate	101	1	1.5838	4.2347	0.9901	44.54	12.85a	3.61%	4.56b	1.46%	4.40b	1.73%	15	1	<1	<1
77	Ethyl heptadecanoate	88	1	14.628	−1.3628	0.9928	6.80	3.98	1.59%	——	——	——	——	——	——	——	——
74	Ethyl oleate	264	1	0.1388	0.4616	0.991	13.51	10.47	3.57%	——	——	——	——	——	——	——	——
78	Ethyl linoleate	67	1	0.4539	−0.0112	0.9954	9.31	103.65	2.10%	——	——	——	——	——	——	——	——
57	Propyl hexanoate	99	1	0.3477	0.0485	0.998	19.55	31.70a	1.21%	1.92b	1.47%	——	——	12.8	2	<1	——
63	Butyl lactate	45	1	1.9302	−0.0979	0.9928	10.03	2.42	1.30%	——	——	——	——	——	——	——	——
56	Hexyl acetate	43	1	1.9779	−0.1351	0.9982	26.36	3.22b	1.75%	——	——	3.35a	1.80%	——	——	——	——
51	Ethyl nonanoate	88	1	1.7241	−0.1371	0.9984	11.19	3.60a	1.63%	3.69a	1.41%	——	——	3.15	1	1	——

Significance: a, b, c: There was significant difference ($p < 0.05$) in the contents of compounds with different letters. The average contents with the letter a were the highest, and the average contents with the letter c were the lowest;

IS: Corresponding internal standard of aroma compounds;

LOD: The lowest concentrations of the analyte standard solutions based on a signal-to-noise ratio of 3;

Odor threshold: were taken from reference. (Liu & Sun, 2018; Fan & Xu, 2011; Wang, Chen, Wu, & Zhao, 2022);

OAV: OAVs were calculated by dividing the concentration by the respective odor threshold;

——: No identified, no found, or no calculated.

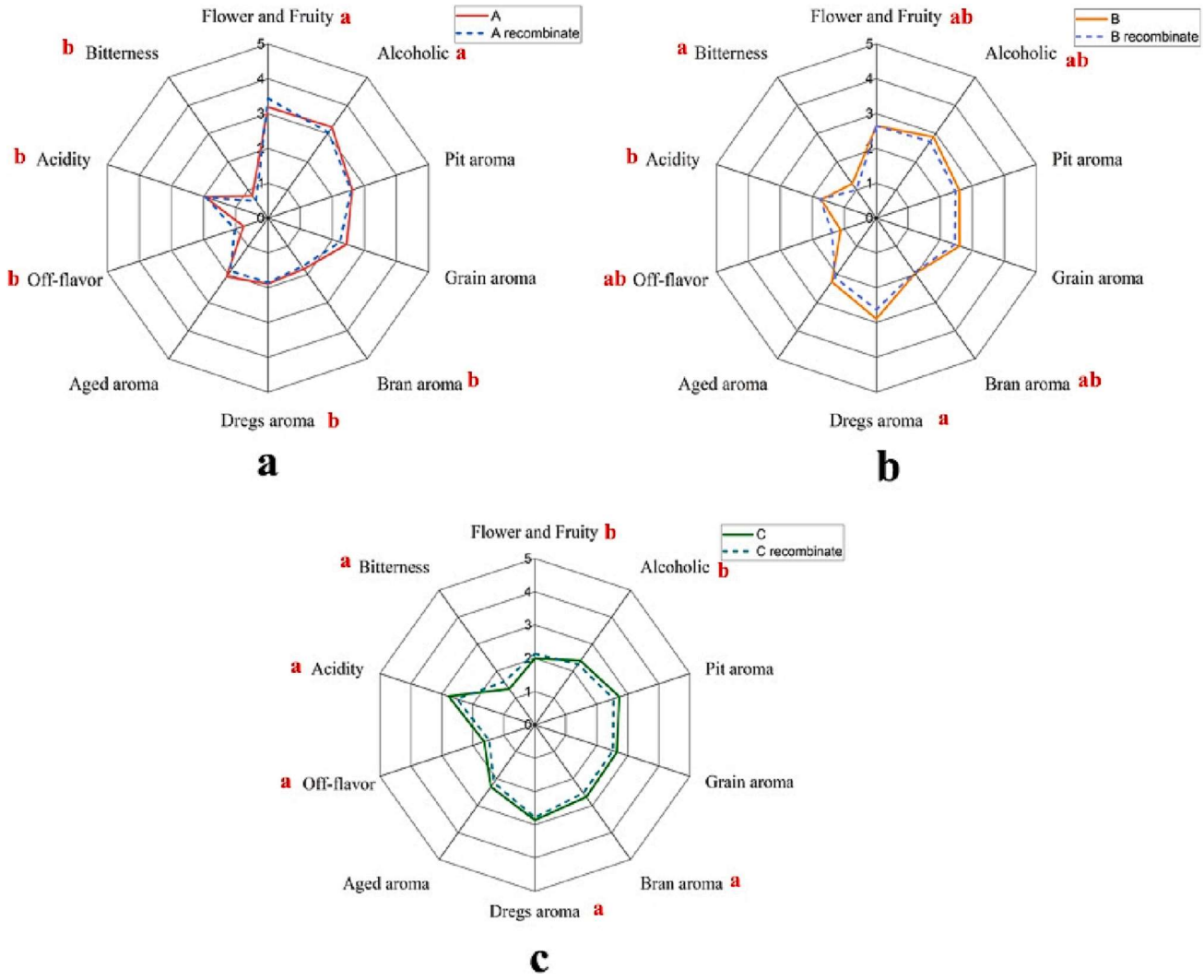


Fig. 1. Aroma recombination of three northern strong aroma type of Baijiu samples. Significance: a, b, c: There was significant difference ($p < 0.05$) in the contents of compounds with different letters. The average contents with a were the highest, and the average contents with c were the lowest.

aroma compounds might be important material basis of the aroma profile of northern strong aroma type of Baijiu. Meanwhile, the compounds with higher FD values also had obvious differences in three samples. For example, the FD value of ethyl hexanoate was the highest in A (6561) and the lowest in C (729), while the FD value of ethyl acetate was the highest in C (2187) and the lowest in A (243). The FD values of pentanoic acid, 3-methylbutanoic acid and ethyl pentanoate in three samples also differed greatly. To sum up, the aroma expression of diverse compounds was unlike, and that of the same compound in different samples was also distinct. It was preliminarily speculated to be one of the molecular factors that caused diversities in flavor quality of northern strong aroma type of Baijiu. Moreover, the specific contribution of aroma compounds to the aroma profile of strong aroma type of Baijiu also needed further study.

3.1.2. *Quantification of aroma compounds*

In order to get a deep insight into the contribution of important aroma compounds on the aroma profile of northern strong aroma type of Baijiu, 68 aroma compounds detected were further quantified by internal standard method combined with GC − MS/SIM. The concentrations of aroma compounds in three samples were analyzed for significance. As shown in Table 2, the concentrations of ethyl hexanoate (1003.24–4506.04 mg/L) and ethyl acetate (1113.45–2824.67 mg/L) were the highest among ester compounds. Meanwhile, the concentrations of 1-butanol (167.03–329.18 mg/L), 3-methyl-1-butanol (140.31–301.36 mg/L) and 1-propanol (243.96–463.67 mg/L) were also high, while acetic acid (121.65–364.29 mg/L) and hexanoic acid (92.56–250.25 mg/L) were more prominent among acid compounds. It was worth noting that the concentrations of aromatic compounds were generally low, and the concentrations of ethyl 3-phenylpropionate and phenylacetaldehyde were relatively high. As shown in the Table 2, the concentrations of aroma compounds in diverse samples were significantly different. For example, ethyl hexanoate was significant in three samples ($\alpha \leq 0.05$) and its concentration was the highest in A (4506.04 mg/L), while the concentration of ethyl acetate was the highest in C (2824.67 mg/L). The concentration of 1-butanol was the highest in A (329.18 mg/L) and the lowest in B (167.03 mg/L), while 3-methyl-1-butanol had no significant difference between A (301.36 mg/L) and B (294.96 mg/L). Hexanoic acid was also significant in three samples ($\alpha \leq 0.05$) and the concentration was the highest in A (250.25 mg/L) and the lowest in C (92.56 mg/L). Moreover, the concentrations of aroma compounds in northern strong aroma type of Baijiu were quite different from that in the other two strong aroma types of Baijiu. For example, the concentration of acids in samples was lower than that in Luzhou Laojiao and Gujinggong, while aromatic compounds were more abundant in northern strong aroma type of Baijiu samples (Zhao et al., 2018; Dong et al., 2019). The ratio of acids to esters has an important impact on the flavor characteristics of strong aroma type of Baijiu. In the national

standard, the ratio of acids to ester was suggested to be above 0.2, while the ratio of acids and esters in northern strong aroma type of Baijiu was low. Acids mainly showed cheesy aroma, while esters mainly represented flower and fruity aroma, and the ratio between them would affect interactive aroma expression. It might be one of the reasons for the formation of unique flavor characteristics of northern strong aroma type of Baijiu. The results also indicated that the concentrations of compounds affected the aroma expression, which might be a notable factor causing the instability in flavor quality of northern strong aroma type of Baijiu.

3.1.3. Oavs of aroma compounds

As described above, the contribution of aroma compounds in Baijiu was not just determined by their concentrations, but also their interactions, as well as matrix effect. Although GC–O analysis is an effective method to identify odorants, it does not consider the matrix effect. Previously studies have reported the ethanol had effect on the perception of aroma compounds. (Sun, 2019; Fan and Xu, 2011; Cao, Li, Tang, Hou, Liu, & Tang, 2017; Zhao et al., 2018) OAVs refers to the ratio of the concentration of aroma compounds to the odor threshold in the corresponding medium. The contribution of aroma compounds to the whole aroma profile of Baijiu could be better understood by calculating the OAVs of aroma compounds. Odor thresholds were taken from previously reports (Liu & Sun, 2018; Fan & Xu, 2011; Wang, Chen, Wu, & Zhao, 2022). As shown in Table 2, a total of 25 aroma compounds were simultaneously confirmed with their OAVs ≥ 1 in all three samples, including 5 alcohols, 6 aromatic compounds, 6 acids, 8 esters. In this context, it was believed that esters were the most important class of aroma compounds of northern strong aroma type of Baijiu owing to their higher OAVs, as well as their corresponding high FD values aforementioned (i.e., ethyl hexanoate, ethyl butanoate, ethyl acetate, etc.). Although the concentrations of aromatic compounds were not high, the OAVs were relatively high because of their low thresholds. Meanwhile, aromatic compounds mainly presented rose fragrance. It was preliminarily speculated that aromatic compounds also have contribution to the aroma profile of northern strong aroma type of Baijiu. Moreover, similar to the previous results, the OAVs of acid compounds in A and B were higher (i.e., butanoic acid, pentanoic acid, hexanoic acid, etc.). It was also speculated that acid compounds were key aroma compounds that affect the flavor quality of northern strong aroma type of Baijiu. According to the determined concentrations as well as FD values, 3-methyl-1-butanol were confirmed as the major aroma compounds, while their OAVs were relatively low. Whether these aroma compounds with strong aroma expression contribute to the overall aroma profile and the interactions between them still needed to be further verified.

3.2. Sensory analysis

3.2.1. Aroma recombination experiments

Based on the research above, aroma compounds with OAVs ≥ 1 were mixed to prepare a model solution in accordance of their concentrations in three samples (Table 2), and used for the following sensory evaluation. The 10 aroma attributes (flower and fruity, alcoholic, pit aroma, grain aroma, bran aroma, dregs aroma, aged aroma, off-flavor, acidity, bitterness) of three sample and its simulated model solution were evaluated by the sensory panelists, and their overall aroma profiles were shown in Fig. 1. As shown in the Fig. 1, the recombinant aroma profile of the simulated model solution had exhibited a good aromatic similarity as that of the original samples (Pearson coefficients were all greater than 0.9). It indicated that the main aroma profile of northern strong aroma type of Baijiu formed with the joint action of these key aroma compounds. Combined with aforementioned results, it can be inferred that the changes of these key compounds might affect the variety of flavor characteristics of northern strong aroma type of Baijiu. However, there was still a slight difference between the recombinant samples and the original samples in few aroma attributes. It could be inferred that except the aforementioned aroma compounds, some volatile compounds also might have a slight impact on these aroma attributes.

Table 3
Omission test from complete reconstitution of aroma compounds.

No.	Aroma compounds omitted from the complete recombinant	n			Significance		
		A	B	C	A	B	C
1	All esters	14	14	14	***	***	***
2	Ethyl hexanoate	14	14	13	***	***	***
3	Ethyl butyrate	13	13	11	***	***	***
4	Ethyl acetate	10	13	14	**	***	***
5	Ethyl lactate	6	4	9			*
6	Ethyl pentanoate	10	10	10	**	**	**
7	All alcohols	4	6	7			
8	3-Methyl-1-butanol	4	4	8			
9	2-Methyl-1-propanol	4	4	6			
10	1-Butanol	6	5	8			
11	All acids	10	10	10	**	**	**
12	Hexanoic acid	10	10	9	**	**	*
13	Butanoic acid	9	9	4	*	*	
14	Acetic acid	5	6	9			*
15	All aromatic compounds	9	9	8	*	*	
16	Phenylacetaldehyde	5	8	4			
17	Ethyl phenylacetate	4	6	4			
18	Ethyl phenylpropionate	9	9	7	*	*	

n: Number of correct judgments from 14 assessors evaluating the aroma difference by the triangle test;
Significance: *, significant ($\alpha \leq 0.05$); **, highly significant ($\alpha \leq 0.01$); ***, very highly significant ($\alpha \leq 0.001$).

In addition, according to the notability analysis based on the sensory attribute scoring of the three samples, the grade differences were found in the flavor characteristics of northern strong aroma of Baijiu. For example, the 'flower and fruity' and 'alcoholic' aroma in A were the strongest, while the 'bran aroma', 'dregs aroma', 'off-flvaor', 'acidity', and 'bitterness' in C were the strongest. Meanwhile, there were no significant differences in pit aroma, grain aroma and aged aroma among the three samples. The previous results showed that aroma compounds were the material basis for the formation of flavor characteristics of Baijiu. However, the influences of key aroma compounds in northern strong aroma type of Baijiu on flavor characteristics still needed to be further analyzed by omission test and flavor matrix.

3.2.2. Aroma omission experiments

To further investigate the contributions of certain aroma compounds to the overall aroma profiles of northern strong aroma type of Baijiu, 18 models with one or a class of selected aroma compounds removed, were subjected to omission experiments. Triangle tests were carried out by the 14 trained sensory panelists. The results of omission test were presented in Table 3. As shown in Table 3, model 1 indicated that most of the evaluators were able to recognize the absence of esters ($p \leq 0.001$), which revealed that these fruity compounds played critical roles in the overall aroma of northern strong aroma type of Baijiu. Among them, a high significance was evaluated when ethyl hexanoate and ethyl butyrate were omitted, followed by ethyl acetate and ethyl pentanoate. Meanwhile, ethyl lactate was calculated for high OAVs (Table 3), but no significant difference was observed in sample A and B. According to previous studies, the concentrations of ethyl acetate and ethyl lactate in strong aroma type of Baijiu needed to be controlled within a certain range to form an appropriate proportion with ethyl hexanoate (Fan, Xu, & Qian, 2019; Xiao, 2022). It was conducive to the formation of typical flavor characteristics of strong aroma type of Baijiu. However, according to the above results, the concentrations and contributions of ethyl acetate and ethyl lactate in different grades of northern strong aroma type of Baijiu were distinct. It was speculated that this was one of the key factors for the unstable flavor quality of northern strong aroma type of Baijiu.

Furthermore, alcohols which were common compounds of northern

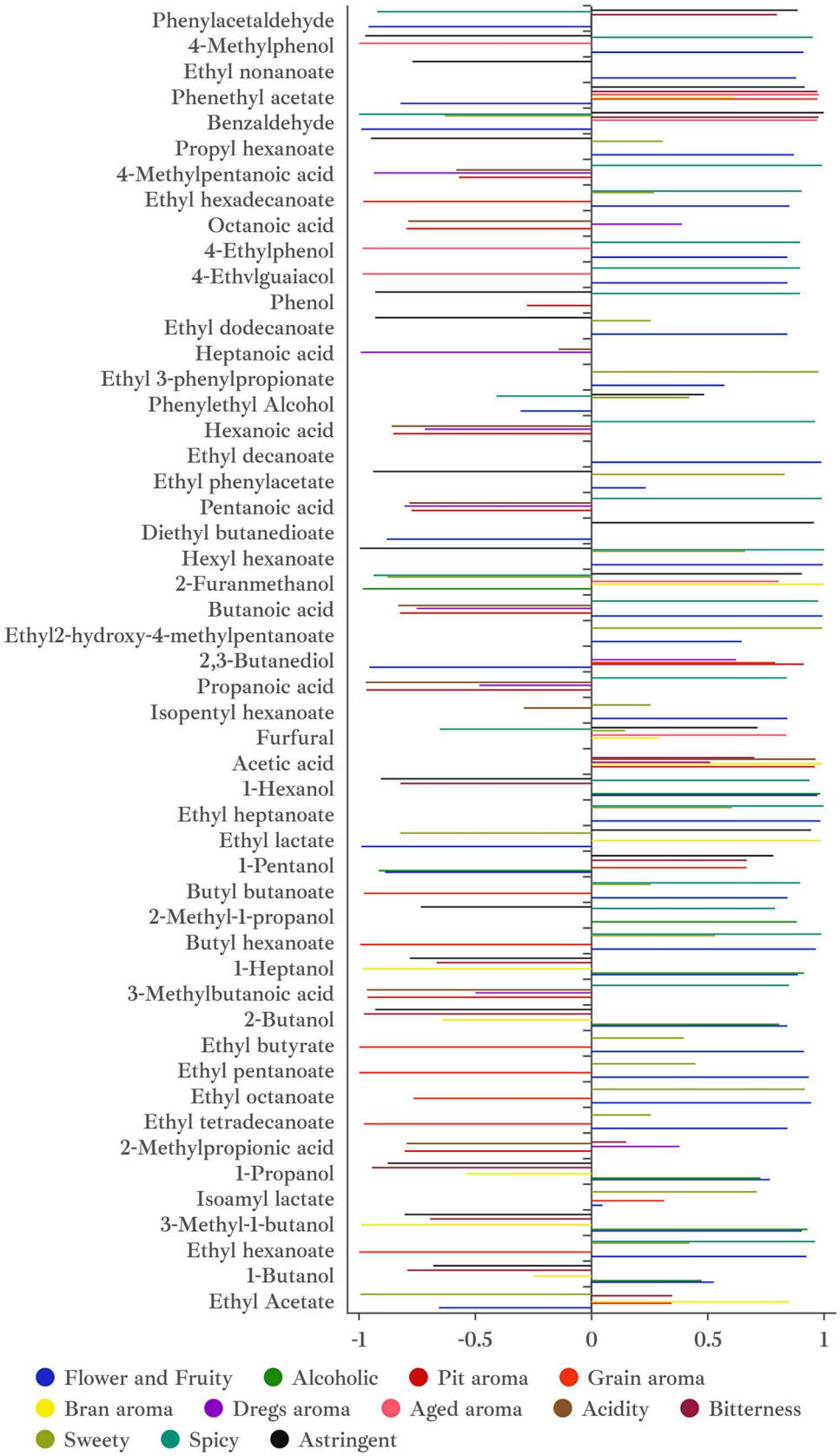


Fig. 2. The relationship between aroma compounds and flavor characteristics based on sensory evaluation.

strong aroma type of Baijiu, were also subjected to the triangle test. However, there were few assessors who could recognize the omission of all alcohols in three samples (in model 7). Therefore, according to the results of the missing test, except ethanol as the basic solvent, alcohols made little contribution to the aroma profile of northern strong aroma type of Baijiu.

With respect to the detected short chain acids, the absences of all acids in model 11 were detected by 10 assessors ($p \leq 0.01$). Further omission tests showed that removal of hexanoic acid (in model 12) could also significantly impact the whole aroma profile of the model, while butanoic acid and acetic acid had different effects in three samples. Specifically, a high significance was shown when butanoic acid was omitted in A and B ($p \leq 0.05$), while the results of acetic acid were opposite. Butanoic acid mainly showed cheesy aroma, while acetic acid was vinegar aroma. The OAVs of butanoic acid and acetic acid were all above 1. However, the contributions of butanoic acid and acetic acid to the aroma profile of three samples were different in the omission test, which might also be the reason for the diversity in flavor quality of northern strong aroma type of Baijiu. Meanwhile, these short chain organic acids were precursors of important ethyl compounds in the brewing process, so the change of acids will also affect the synthesis of some ester compounds.

As a rich variety of compounds in northern strong aroma type of Baijiu, the concentrations of aromatic compounds were generally low, but the OAVs of some compounds were still high because of low thresholds. However, in the omission test, only 9 out of 14 assessors could recognize the omission of all aromatic compounds (in model 15) and ethyl phenylpropionate (in model 18) in A and B. Hence, it was speculated that ethyl phenylpropionate contributed more to the aroma profile of northern strong aroma type of Baijiu among the aromatic compounds.

In general, ethyl hexanoate, ethyl butyrate, ethyl acetate, ethyl pentanoate, hexanoic acid, butanoic acid and ethyl phenylpropionate were confirmed as key aroma compounds in northern strong aroma type of Baijiu, which was similar to other strong aroma type of Baijiu (He et al., 2022; He, Liu, Qian, Yu, Xu, & Chen, 2020). Moreover, the different aroma expressions of some compounds (i.e., ethyl lactate, hexanoic acid, acetic acid, ethyl phenylpropionate) might affect the stability of flavor quality of northern strong aroma type of Baijiu.

3.2.3. Correlation analysis of aroma compounds and aroma profile of strong aroma type of Baijiu by flavor matrix

To deeply confirm the relative contributions of aroma compounds to flavor characteristics of northern strong aroma type of Baijiu, 12 flavor characteristics of three sample (flower and fruity, alcoholic, pit aroma, grain aroma, bran aroma, dregs aroma, aged aroma, acidity, bitterness, sweety, spicy, astringent) were scored 0–5 by assessors. Then the Pearson correlation coefficient between the OAVs of aroma compounds and the scores was calculated and plotted based on it. As shown in the Fig. 2, most alcohols had positive correlations with flower and fruity and alcoholic, but negative correlations with bran aroma, bitterness and astringent aroma. In particular, 1-pentanol was negatively correlated with flower and fruity and alcoholic, while positively correlated with the grain aroma, bitterness and astringent aroma. Meanwhile, 2,3-butanediol was positively correlated with pit aroma, grain aroma and dregs aroma. As described above, alcohols had little effect on the overall aroma profile, but various alcohols still contributed to the different flavor characteristics of northern strong aroma type of Baijiu.

Most acids were negatively correlated with pit aroma and dregs aroma, and positively correlated with spicy. Among them, butanoic acid had a strong positive correlation with flower and fruity and spicy, and hexanoic acid also had a strong positive correlation with spicy. Significantly, apart from their effects on the flower and fruity and sweety, some of aromatic compounds were also positively correlated with spicy, such as phenol, 4-ethylguaiacol, 4-ethylphenol, 4-methylphenol.

As the most abundant class of compounds, most esters had a positive correlation with flower and fruity and sweety. Meanwhile, some esters had a strong positive correlation with spicy taste, such as ethyl hexanoate, butyl hexanoate, butyl butanoate, ethyl heptanoate, hexyl hexanoate, and ethyl hexadecanoate. It was preliminarily indicated that they had a certain contribution to the spicy taste of northern strong aroma type of Baijiu, which was similar to the previous research (He, Tang, Yu, Chen, & Xu, 2022).

On the whole, the flavor characteristics of northern strong aroma type of Baijiu were formed by the combined action of various aroma compounds. For example, most alcohols, aromatic compounds, and esters with high FD values and OAVs had a positive impact on flower and fruity aroma, while alcoholic was mainly affected by alcohols. The spicy taste was influenced by a combination of some alcohols, aromatics, acids, and esters, and astringency was mainly affected by alcohols, aromatic compounds, and some esters. Additionally, acetic acid contributed to pit aroma, whereas acetic acid, ethyl acetate, and ethyl lactate with high FD values and OAVs had positive impacts on bran aroma. Based on the aforementioned results, the unique aroma profile of northern strong aroma type of Baijiu was formed by the key aroma compounds with different aroma expression.

4. Conclusion

In summary, this study identified a total of 78 aroma compounds of northern strong aroma type of Baijiu, which were identified by GC–O and GC–MS. Among them, 25 aroma compounds were further recognized as the important aroma compounds owing to their relatively high OAVs in all samples. Furthermore, the aroma profile of northern strong aroma type of Baijiu was successfully simulated based on the aforementioned aroma compounds (OAVs $\geq$ 1). The omission test proved ethyl hexanoate, ethyl butyrate, ethyl acetate, ethyl pentanoate, hexanoic acid, butanoic acid and ethyl phenylpropionate as the key aroma compounds. Then the relationship between aroma compounds and 12 flavor characteristics was calculated and illustrated. As a result, various aroma compounds had different degrees of influence on the flavor characteristics of northern strong aroma type of Baijiu. Hence, the overall aroma profile of northern strong aroma type of Baijiu was formed by the key aroma compounds with different aroma expression. Meanwhile, the different aroma expressions of these aroma compounds might affect the stability of flavor quality of northern strong aroma type of Baijiu.

CRediT authorship contribution statement

Jiaxin Hong: Conceptualization, Investigation, Methodology, Data curation, Writing – original draft, Writing – review & editing. **He Huang:** Software, Visualization. **Dongrui Zhao:** Supervision, Conceptualization, Writing – review & editing. **Jinyuan Sun:** Formal analysis. **Mingquan Huang:** Formal analysis. **Xiaotao Sun:** Resources. **Baoguo Sun:** Resources, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

Chen, P. P., Liu, Y., Wu, J. H., Yu, B., Zhao, H., Huang, M. Q., & Zheng, F. P. (2022a). Sensory-directed decoding of key aroma compounds from Jiugui-series Baijiu, the representative of Fuyu-flavor-type Baijiu (FFTB). *Journal of Food Composition and Analysis, 114*, Article 104799. https://doi.org/10.1016/j.jfca.2022.104799

Chen, H., Wu, Y. S., Wang, J. S., Hong, J. X., Tian, W. J., Zhao, D. R., … Sun, B. G. (2022b). Uncover the flavor code of roasted sesame for sesame flavor Baijiu: advance on the revelation of aroma compounds in sesame flavor baijiu by means of modern separation technology and molecular sensory evaluation. *Foods, 11*(7), 998–1020. https://doi.org/10.3390/foods11070998

Cao, R. J., Li, A. J., Tang, Y. H., Hou, S. L., Liu, G. Y., & Tang, Z. H. (2017). Analysis of flavoring components of base liquor produced in pits of different age based on multivariate statistical analysis and OAV. *Liquor-Making Sci. Technol., 11*, 44–50. https://doi.org/10.13746/j.njkj.2017205

Dong, W., Guo, R. N., Liu, M., Shen, C. H., Sun, X. T., Zhao, M. M., … Wu, J. H. (2019). Characterization of key odorants causing the roasted and mud-like aromas in strong-aroma types of base Baijiu. *Food Research International., 125*, Article 108546. https://doi.org/10.1016/j.foodres.2019.108546

Fan, W. L., & Xu, Y. (2011). Determination of odor thresholds of volatile aroma compounds in baijiu by a forced-choice ascending concentration series method of limits. *Liquor Making, 38*(4), 80–84. https://doi.org/10.3969/j.issn.1002-8110.2011.04.034

Fan, W. L., & Xu, Y. (2014). Current practice and future trends of aroma and flavor of Chinese liquor (baijiu). *Journal of Food Safety and Quality, 5*(10), 3073–3078. https://doi.org/10.19812/j.cnki.jfsq11-5956/ts.2014.10.019

Fan, W., & Qian, M. (2005). Headspace solid phase microextraction and gas chromatography−olfactometry dilution analysis of young and aged chinese "yanghe daqu" liquors. *Journal of Agricultural and Food Chemistry, 53*, 7931–7938. https://doi.org/10.1021/jf051011k

Fan, W., Xu, Y., & Qian, M. (2019). Current practice and future trends of aroma and flavor research in Chinese Baijiu. In B. Guthrie, J. D. Beauchamp, A. Buettner, S. Toth, & M. C. Qian (Eds.), *Sex, smoke, and spirits: The role of chemistry* (pp. 145–175). Washington: American Chemical Society (ACS). https://pubs.acs.org/doi/abs/10.1021/bk-2019-1321.ch012.

Gao, W. J., Fan, W. L., & Xu, Y. (2014). Characterization of the key odorants in light aroma type Chinese liquor by gas chromatography-olfactometry, quantitative measurements, aroma recombination, and omission studies. *Journal of Agricultural and Food Chemistry, 62*, 5796–5804. https://doi.org/10.1021/jf501214c

Hong, J. X., Zhao, D. R., & Sun, B. G. (2021). Research progress on the profile of trace components in baijiu. *Food Reviews International, Online.*. https://doi.org/10.1080/87559129.2021.1936001

He, Y. X., Liu, Z. P., Qian, M., Yu, X. W., Xu, Y., & Chen, S. (2020). Unraveling the chemosensory characteristics of strong-aroma type Baijiu from different regions using comprehensive twodimensional gas chromatography-time-of-flight mass spectrometry and descriptive sensory analysis. *Food Chemistry, 331*, Article 127335. https://doi.org/10.1016/j.foodchem.2020.127335

Hong, J. X., Wang, J. S., Zhang, C. S., Zhao, Z. G., Tian, W. J., Wu, Y. S., … Sun, J. Y. (2021). Unraveling variation on the profile aroma compounds of strong aroma type of Baijiu in different regions by molecular matrix analysis and olfactory analysis. *RSC Advances, 11*, 33511–33521. https://doi.org/10.1039/D1RA90161C

He, F., Yang, S. Q., Zhang, G. H., Xu, L., Li, H. H., Sun, J. Y., … Sun, B. G. (2022). Exploration of key aroma active compounds in strong flavor Baijiu during the distillation by modern instrument detection technology combined with multivariate statistical analysis methods. *Journal of Food Composition and Analysis, 110*, Article 104577. https://doi.org/10.1016/j.jfca.2022.104577

He, Y. X., Tang, K., Yu, X. W., Chen, S., & Xu, Y. (2022). Identification of compounds contributing to trigeminal pungency of baijiu by sensory evaluation, quantitative measurements, correlation analysis, and sensory verification testing. *Journal of Agricultural and Food Chemistry, 70*(2), 598–606. https://doi.org/10.1021/acs.jafc.1c06875

Hu, X. L., Du, H., & Xu, Y. (2015). Identification and quantification of the caproic acid-producing bacterium Clostridium kluyveri in the fermentation of pit mud used for Chinese strong-aroma type liquor production. *International Journal of Food Microbiology, 214*, 116–122. https://doi.org/10.1016/j.ijfoodmicro.2015.07.032

Liu, H. L., & Sun, B. G. (2018). Effect of fermentation processing on the flavor of baijiu. *Journal of Agricultural and Food Chemistry, 66*, 5425–5432. https://doi.org/10.1021/acs.jafc.8b00692

Liu, M. K., Tang, Y. M., Zhao, K., Liu, Y., Guo, X. J., Ren, D. Q., … Zhang, X. P. (2017). Determination of the fungal community of pit mud in fermentation cellars for Chinese strong-flavor liquor, using DGGE and Illumina MiSeq sequencing. *Food Research International, 91*, 80–87. https://doi.org/10.1016/j.foodres.2016.11.037

Mao, F. J., Huang, J., Zhou, R. Q., Qin, H., Zhang, S., Cai, X., & Qiu, C. (2022). Effects of different daqu on microbial community domestication and metabolites in nongxiang baijiu brewing microecosystem. *Front. Microbiol., 13*, Article 939904. https://doi.org/10.3389/fmicb.2022.939904

Song, X., Si, J., Zhu, L., et al. (2020). Untargeted and targeted metabolomics strategy for the classification of strong aroma-type baijiu (liquor) according to geographical origin using comprehensive two-dimensional gas chromatography-time-of-flight mass spectrometry[J]. *Food Chemistry, 134*, 126098.

Sun, B. G. (2019). *Baijiu and huangjiu-chinese national alcohols* (pp. 109–115). Chemical Industry: Beijing, China.

Sun, J., Zhao, D., Zhang, F., Sun, B., Zheng, F., … Huang, M. (2017). Joint direct injection and GC-MS chemometric approach for chemical profile and sulfur compounds of sesame flavor Chinese Baijiu (Chinese liquor) [J]. *European Food Research & Technology, 244*(9), 1–16.

Wang, J. S., Chen, H., Wu, Y. S., & Zhao, D. R. (2022). Uncover the flavor code of strong-aroma baijiu: Research progress on the revelation of aroma compounds in strong-aroma baijiu by means of modern separation technology and molecular sensory evaluation. *Journal of Food Composition and Analysis, 109*, 104499–104516. https://doi.org/10.1016/j.jfca.2022.104499

Wang, X. Q., Guo, W. T., Sun, B. G., Li, H. H., Zheng, F. P., Li, J. C., & Meng, N. (2022). Characterization of key aroma-active compounds in two types of peach spirits produced by distillation and pervaporation by means of the sensomics approach. *Foods, 11*, 2598. https://doi.org/10.3390/foods11172598

Wu, Y. S., Chen, H., Huang, H., Chen, F. Y., Hong, J. X., Zhao, D. R., … Sun, B. G. (2023). Revelation for the influence mechanism of long-chain fatty acid ethyl esters on the baijiu quality by multicomponent chemometrics combined with modern flavor sensomics. *Foods, 12*, 1267. https://doi.org/10.3390/foods12061267

Xiao, D. G. (2022). Discussion on the formation mechanism of ester compounds in the production process of baijiu. *Liquor-making Science & Technology, 9*, 17–24. https://doi.org/10.13746/j.njkj.2022161

Xu, Y. Q., Wu, M. Q., Niu, J. L., Lin, M. W., Zhu, H., Wang, K., … Sun, B. G. (2023). Characteristics and correlation of the microbial communities and flavor compounds during the first three rounds of fermentation in Chinese sauce-flavor Baijiu. *Foods, 12*(1), 207. https://doi.org/10.3390/foods12010207

Yu, Y. M., Nie, Y., Chen, S., & Xu, Y. (2023). Characterization of the dynamic retronasal aroma perception and oral aroma release of Baijiu by progressive profiling and an intra-oral SPME combined with GC × GC-TOFMS method. *Food Chemistry, 405*, Article 134854. https://doi.org/10.1016/j.foodchem.2022.134854

Zhao, D. R., Shi, D. M., Sun, J. Y., Li, A. J., Sun, B. G., Zhao, M. M., … Zheng, F. P. (2018). Characterization of key aroma compounds in Gujinggong Chinese Baijiu by gas chromatography-olfactometry, quantitative measurements, and sensory evaluation. *Food Research International, 105*, 616–627. https://doi.org/10.1016/j.foodres.2017.11.074

Zhu, L., Wang, X. L., Song, X. B., Zheng, F. P., Li, H. H., Chen, F., … Zhang, F. Y. (2020). Evolution of the key odorants and aroma profiles in traditional Laowuzeng baijiu during its one-year ageing. *Food Chemistry, 310*, Article 125898. https://doi.org/10.1016/j.foodchem.2019.125898

Further reading

Li, H. H. (2018). Joint direct injection and GC-MS chemometric approach for chemical profile and sulfur compounds of sesame flavor Chinese Baijiu (Chinese liquor). *European Food Research and Technology, 244*(1), 145–160. https://doi.org/10.1007/s00217-017-2938-7

Sun, J. Y., Zhao, D. R., Zhang, F. G., Sun, B. G., Zheng, F. P., Huang, M. Q., Sun, X. T., Song, X. B., Jing, S., Zhu, L., Ma, C. F., Song, T., Wu, J. H., Zhao, Q. Z., Zheng, F. P., Zhao, M. M., & Chen, F. (2020). Untargeted and targeted metabolomics strategy for the classification of strong aroma-type baijiu (liquor) according to geographical origin using comprehensive two dimensional gas chromatography-time-of-flight mass spectrometry. *Food Chemistry, 314*. https://doi.org/10.1016/j.foodchem.2019.126098